Ke Zhang

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Research Interests

Planet formation, protoplanetary disks, astrochemistry, thermo-chemical modeling of disks, (sub)millimeter interferometry, infrared spectroscopy (JWST and ground-based)

Selected Highlights

- **Publications:** 95+ refereed papers · H-index: 40 · 5,400+ citations
- **Talks:** 30+ invited talks, colloquia, and seminars
- **Research Funding:** $2.5M+ awarded as PI (JWST, NSF, NASA XRP, NASA Hubble Fellowship)
- **Major Observing Programs:** PI of ALMA Large Program AGE-PRO (103h) · PI of 2 JWST programs (133h)
- **Survey Leadership:** PI group member of JDISC (JWST Disk Infrared Spectral Chemistry Survey)
- **Mentoring:** 3 PhD students, including 2 NSF GRFP awardees · 3 postdocs

Education

California Institute of Technology — Pasadena, CA USA
Ph.D. in Astrophysics. Advisor: Geoffrey A. Blake — 10/2009–06/2015
Thesis: Volatiles in Protoplanetary Disks

Beijing Normal University — Beijing, CHINA
M.S. in Astrophysics — 09/2005–07/2008
B.S. in Astronomy — 09/2001–07/2005

Professional Employment

Associate Professor — Madison, WI USA
Department of Astronomy, University of Wisconsin-Madison — 08/2026–Present

Assistant Professor — Madison, WI USA
Department of Astronomy, University of Wisconsin-Madison — 08/2020–07/2026

NASA Hubble Fellow — Ann Arbor, MI USA
Department of Astronomy, University of Michigan — 08/2017–07/2020

Postdoctoral Researcher — Ann Arbor, MI USA
Department of Astronomy, University of Michigan — 07/2015–07/2017

Awards and Honors

- Scialog Fellow: Signatures of Life in the Universe 03/2020
- NASA Hubble Fellowship 01/2017
- NASA Sagan Fellowship (declined) 01/2017
- The ALMA Ambassadors Postdoctoral Program 01/2017
- Chinese Government Award for Outstanding Self-financed Students Abroad 03/2015

Group Honors (students and postdocs advised)

- NSF Astronomy and Astrophysics Postdoctoral Fellowship: Abygail Waggoner (2026)
- NSF Graduate Research Fellowship: Estephani TorresVillanueva (2023), Tayt Armitage (2025)
- UW-Madison Sophomore Research Fellowship: Vicki Braianova (2022), Ella Chevalier (2023), Nick Tallon (2025), Carl Yao (2025)
- Wisconsin Space Consortium Undergraduate Research Award: Nina Weichmann (2023), Nick Tallon (2025)

External Research Support

Current – Major Research Grants

2026–2029 Dynamic Disks and Evolving Chemistry – Tracing the Origins of Planetary Volatile Composition
NASA Exoplanets Research Program (XRP) · **PI** · $451k

2026–2028 Population Study of Chemistry in the First Million Years of Planet Formation: A MIRI Survey of Embedded Disks in Ophiuchus
STScI (JWST GO4) · **PI** · $380k total ($280k to UW-Madison)

2026–2029 Integrating Pebble Drift with Disk Chemistry: A New Framework for Interpreting MIRI Water Lines
STScI (JWST GO5) · co-PI · $50k to UW-Madison

2024–2026 Building on ALMA: A JWST Legacy Survey of the Chemical Evolution of Planet-Forming Disks
STScI (JWST GO2) · **PI** · $808k total ($412k to UW-Madison)

Current – Student Fellowships

2026–2029 NSF Graduate Research Fellowship – Tayt Armitage
National Science Foundation · advisor · $147k

2023–2026 NSF Graduate Research Fellowship – Estephani TorresVillanueva
National Science Foundation · advisor · $147k

Past

2025–2026 Dynamic Evolution of Water Reservoirs in Planet-Forming Disks
UW-Madison Fall Competition · **PI** · $60k

2024–2026 Searching for Signs of Life in Exoplanet Hycean Worlds
Wisconsin Alumni Research Foundation (WARF) · co-I · $50k (Zhang share)

2023–2026 A DSHARP-MIRI Treasury Survey of Chemistry in Planet-Forming Regions
STScI (JWST GO1) · co-I · $36k to UW-Madison

2022–2025 The Evolution of Gas in Protoplanetary Disks
NSF Astronomy and Astrophysics Research Grants · **PI** · $444k

2022–2023 NRAO Student Observing Support Award
National Radio Astronomy Observatory · **PI** · $31k

2017–2020 NASA Hubble Fellowship
NASA / STScI · **PI** · $330k

2017–2018 The ALMA Ambassador
National Radio Astronomy Observatory · **PI** · $10k

Accepted Observing Proposals

Proposals led by my group

Total: ALMA 230.8h · JWST 133h · NOEMA 67.4h · VLA 7.9h

2026 How Much Gas Survives? Probing the Final Reservoir in 6–8 Myr Protoplanetary Disks
ALMA Cycle 13 · **PI** · B grade · 31.5h

2026 The First Millimeter Census of Disks around Stars below 0.1 $M_\odot$ at 5–15 Myr
ALMA Cycle 13 · **PI** · C grade · 26.7h

2026 Integrating Pebble Drift with Disk Chemistry: A New Framework for Interpreting MIRI Water Lines
JWST GO5 Theory · PI: J. Calahan, **co-PI**

2025 Population Study of Chemistry in the First Million Years of Planet Formation: A MIRI Survey of Embedded Disks in Ophiuchus
JWST GO4 · **PI** · 68h

2023 Measuring accurate gas masses for the exoALMA planet-forming disks
ALMA Cycle 10 · PI: L. Trapman (postdoc) · B grade · 13.6h

2023 Exploring extreme pebble drift and growth in the HD 179218 protoplanetary disk
VLA 24A · PI: L. Trapman (postdoc) · A grade · 7.9h

2023 Building on ALMA: A JWST Legacy Survey of the Chemical Evolution of Planet-Forming Disks
JWST GO2 · **PI** · 65h

2022 A unique gas tracer of pebble drift in protoplanetary disks
ALMA Cycle 9 · **PI** · A grade · 9h

2022 Weighing the Elias 2-27 protoplanetary disk
ALMA Cycle 9 · PI: L. Trapman (postdoc) · A grade · 3.5h

2022 Measuring accurate gas masses of the planet-forming disks in Lupus
ALMA Cycle 9 · PI: L. Trapman (postdoc) · B grade · 26.5h

2021 AGE-PRO: The ALMA Survey of Gas Evolution in PROtoplanetary Disks
ALMA Cycle 8 **Large Program** · **PI** · A grade · 103h

2021 A unique gas tracer of pebble drift in protoplanetary disks
ALMA Cycle 8 · **PI** · B grade · 9h

2021 The first robust mass measurement of a Herbig disk
NOEMA · **PI** · B grade · 10h

2020 CO as a Tracer of Pebble Growth and Drift in Protoplanetary disks
NOEMA · **PI** · A grade · 20h

2019 The Chemistry of Planet Formation within the First Myr
NOEMA · **PI** · A grade · 16h

2018 Gas mass in circumstellar disks younger than 1 Myr
ALMA Cycle 6 · **PI** · C grade · 8h

2017 Does CO depletion happen within the first Myr of disk formation?
NOEMA · **PI** · B grade · 9h

2017 Gas mass distribution in the HD 163296 protoplanetary disk
NOEMA · **PI** · A grade · 12.4h

Co-I proposals (summary)

JWST: 7 programs (113h total) · **ALMA:** 15+ programs including 1 Large Program (130h)

Publications

- Total career citations recorded in SciX (NASA Science Explorer, successor to ADS): 5474
  link to publication list on SciX
- H-index: 40
- Top five first-author publications:
  Zhang et al. 2015 (363 citations), Zhang et al. 2021 (177 citations), Zhang et al. 2017 (136 citations), Zhang et al. 2016 (106 citations), Zhang et al. 2019 (105 citations)

First-author papers and papers led by students/postdocs advised

⋆: postdoc advised, *o*: student advised

27. **Zhang, K.**, Banzatti, A., Salyk, C., Waggoner, A., Pontoppidan, K., Colmenares, M., & et al. (2026), The JDISC Survey: Inner Disk Chemistry of Class I/FS Disks and Tentative Evidence for Early Pebble Drift, AJ, 172, 156.

26. ^{o}Armitage, T., Williams, J., **Zhang, K.**, Krijt, S., Trapman, L., Booth, R., & et al. (2026), Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement, ApJ, 998, 308.

25. ^{o}Yao, J., **Zhang, K.**, Banzatti, A., Bajaj, N., Pascucci, I., Miley, J., & et al. (2026), Molecular Similarity and Water Diversity in Coeval Binary Disks: JWST/MIRI Observations of Sz 65 and Sz 66, ApJ, 1005, 70.

24. ^{o}Chevalier, E., **Zhang, K.**, Vioque, M., Kurtovic, N., Pinilla, P., Miley, J., & et al. (2026), The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO): Formaldehyde (H_2CO) Emission and Its Links to Disk Properties, ApJ, 1005, 199.

23. ⋆Trapman, L., Longarini, C., Rosotti, G., Andrews, S., Bae, J., Barraza-Alfaro, M., & et al. (2025),

exoALMA. XIII. Gas Masses from N_2H^+ and $C^{18}O$: A Comparison of Measurement Techniques for Protoplanetary Gas Disk Masses, ApJL, 984, L18.

22. **Zhang, K.**, Pérez, L., Pascucci, I., Pinilla, P., Cieza, L., Carpenter, J., & et al. (2025), The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). I. Program Overview and Summary of First Results, ApJ, 989, 1.

21. *Trapman, L., **Zhang, K.**, Rosotti, G., Pinilla, P., Tabone, B., Pascucci, I., & et al. (2025), The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). V. Protoplanetary Gas Disk Masses, ApJ, 989, 5.

20. *Trapman, L., Vioque, M., Kurtovic, N., **Zhang, K.**, Rosotti, G., Pinilla, P., & et al. (2025), The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). XI. Beam-corrected Gas Disk Sizes from Fitting ^{12}CO Moment Zero Maps, ApJ, 989, 10.

19. Zhang, S., ^{o}Kalscheur, M., Long, F., **Zhang, K.**, Long, D., Bergin, E., & et al. (2023), Substructures in Compact Disks of the Taurus Star-forming Region, ApJ, 952, 108.

18. *Trapman, L., Rosotti, G., **Zhang, K.**, & Tabone, B. (2023), How Large Is a Disk-What Do Protoplanetary Disk Gas Sizes Really Mean?, ApJ, 954, 41.

17. *Trapman, L., Tabone, B., Rosotti, G., & **Zhang, K.** (2022), Effect of MHD Wind-driven Disk Evolution on the Observed Sizes of Protoplanetary Disks, ApJ, 926, 61.

16. *Trapman, L., **Zhang, K.**, van't Hoff, M., Hogerheijde, M., & Bergin, E. (2022), A Novel Way of Measuring the Gas Disk Mass of Protoplanetary Disks Using N_2H^+ and $C^{18}O$, ApJL, 926, L2.

15. **Zhang, K.**, Booth, A., Law, C., Bosman, A., Schwarz, K., Bergin, E., & et al. (2021), Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions, ApJS, 257, 5.

14. **Zhang, K.**, Bosman, A., & Bergin, E. (2020), Excess C/H in Protoplanetary Disk Gas from Icy Pebble Drift Across the CO Snowline, ApJL, 891, L16.

13. **Zhang, K.**, Schwarz, K., & Bergin, E. (2020), Rapid Evolution of Volatile CO from the Protostellar Disk Stage to the Protoplanetary Disk Stage, ApJL, 891, L17.

12. ^{o}Long, D., **Zhang, K.**, Teague, R., & Bergin, E. (2020), Hints of a Population of Solar System Analog Planets from ALMA, ApJL, 895, L46.

11. **Zhang, K.**, Bergin, E., Schwarz, K., Krijt, S., & Ciesla, F. (2019), Systematic Variations of CO Gas Abundance with Radius in Gas-rich Protoplanetary Disks, ApJ, 883, 98.

10. **Zhang, K.**, Bergin, E., Blake, G., Cleeves, L., & Schwarz, K. (2017), Mass inventory of the giant-planet formation zone in a solar nebula analogue, Nature Astronomy, 1, 0130.

9. **Zhang, K.**, Bergin, E., Blake, G., Cleeves, L., Hogerheijde, M., Salinas, V., & et al. (2016), On the Commonality of 10-30 AU Sized Axisymmetric Dust Structures in Protoplanetary Disks, ApJL, 818, L16.

8. **Zhang, K.**, Crockett, N., Salyk, C., Pontoppidan, K., Turner, N., Carpenter, J., & et al. (2015), Dimming and CO absorption toward the AA Tau protoplanetary disk: An infalling flow caused by disk instability?, ApJ, 805, 55.

7. **Zhang, K.**, Blake, G., & Bergin, E. (2015), Evidence of Fast Pebble Growth Near Condensation Fronts in the HL Tau Protoplanetary Disk, ApJL, 806, L7.

6. **Zhang, K.**, Isella, A., Carpenter, J., & Blake, G. (2014), Comparison of the Dust and Gas Radial Structure in the Transition Disk [PZ99] J160421.7-213028, ApJ, 791, 42.

5. **Zhang, K.**, Pontoppidan, K., Salyk, C., & Blake, G. (2013), Evidence for a Snow Line beyond the Transitional Radius in the TW Hya Protoplanetary Disk, ApJ, 766, 82.

4. **Zhang, K.**, Jiang, B., & Li, A. (2009), On Magnesium Sulfide as the Carrier of the 30 μm Emission Feature in Evolved Stars, ApJ, 702, 680.

3. **Zhang, K.**, Jiang, B., & Li, A. (2009), On the carriers of the 21μm emission feature in post-asymptotic giant branch stars, MNRAS, 396, 1247.

2. **Zhang, K.**, & Jiang, B. (2008), The infrared spectral features of circumstellar envelope of evolved low- and intermediate-mass stars, Science in China: Physics, Mechanics and Astronomy, 51, 1187.

1. **Zhang, K.**, Jiang, B., & Li, A. (2006), The 21 micron feature in the circumstellar envelopes around highly evolved stars, Progress in Astronomy, 24, 43.

Book Chapters

Zhang, K., 2024, Chemistry in Protoplanetary Disks, Chapter 2 in the Reviews in Mineralogy and Geochemistry (RiMG) Volume 90 on "Exoplanets: Compositions, Mineralogy, and Evolution"

Co-author publications

Co-author on 71 additional papers (listed below without titles); full list with titles available on SciX.

71. Law, C., Le Gal, R., et al., incl. **Zhang, K.** (2026), ApJ, 997, 91.
70. Mallaney, P., Banzatti, A., et al., incl. **Zhang, K.** (2026), ApJ, 998, 255.
69. Salyk, C., Pontoppidan, K., **Zhang, K.**, et al. (2026), ApJ, 999, 92.
68. Jiang, S., Huang, J., et al., incl. **Zhang, K.** (2026), ApJ, 1000, 35.
67. Salyk, C., Pontoppidan, K., et al., incl. **Zhang, K.** (2025), AJ, 169, 184.
66. Arulanantham, N., Salyk, C., et al., incl. **Zhang, K.** (2025), AJ, 170, 67.
65. Long, F., Pascucci, I., et al., incl. **Zhang, K.** (2025), ApJL, 978, L30.
64. Law, C., Le Gal, R., et al., incl. **Zhang, K.** (2025), ApJ, 985, 84.
63. Ruiz-Rodriguez, D., González-Ruilova, C., et al., incl. **Zhang, K.** (2025), ApJ, 989, 2.
62. Deng, D., Vioque, M., et al., incl. **Zhang, K.** (2025), ApJ, 989, 3.
61. Agurto-Gangas, C., Pérez, L., et al., incl. **Zhang, K.** (2025), ApJ, 989, 4.
60. Kurtovic, N., Gárate, M., et al., incl. **Zhang, K.** (2025), ApJ, 989, 6.
59. Tabone, B., Rosotti, G., et al., incl. **Zhang, K.** (2025), ApJ, 989, 7.
58. Anania, R., Rosotti, G., et al., incl. **Zhang, K.** (2025), ApJ, 989, 8.

57. Vioque, M., Kurtovic, N., et al., incl. **Zhang, K.** (2025), ApJ, 989, 9.
56. Miley, J., Pérez, L., et al., incl. **Zhang, K.** (2025), ApJ, 989, 11.
55. Krijt, S., Banzatti, A., **Zhang, K.**, et al. (2025), ApJL, 990, L72.
54. Houge, A., Krijt, S., et al., incl. **Zhang, K.** (2025), MNRAS, 537, 691.
53. Pinilla, P., Sierra, A., et al., incl. **Zhang, K.** (2025), MNRAS, 543, 2723.
52. Aloisi, R., Vanderburg, A., et al., incl. **Zhang, K.** (2025), PASP, 137, 114404.
51. Pontoppidan, K., Salyk, C., et al., incl. **Zhang, K.** (2024), ApJ, 963, 158.
50. Bergin, E., Bosman, A., et al., incl. **Zhang, K.** (2024), ApJ, 965, 147.
49. Anderson, D., Cleeves, L., et al., incl. **Zhang, K.** (2024), ApJ, 966, 84.
48. Calahan, J., Bergin, E., et al., incl. **Zhang, K.** (2024), ApJ, 967, 158.
47. Sierra, A., Pérez, L., et al., incl. **Zhang, K.** (2024), ApJ, 974, 102.
46. Calahan, J., Bergin, E., et al., incl. **Zhang, K.** (2024), ApJ, 975, 170.
45. Garrido-Deutelmoser, J., Petrovich, C., et al., incl. **Zhang, K.** (2023), ApJL, 945, L37.
44. Banzatti, A., Pontoppidan, K., et al., incl. **Zhang, K.** (2023), ApJL, 957, L22.
43. Xie, C., Pascucci, I., et al., incl. **Zhang, K.** (2023), ApJL, 959, L25.
42. Calahan, J., Bergin, E., et al., incl. **Zhang, K.** (2023), Nat. Astron., 7, 49.
41. Anderson, D., Cleeves, L., et al., incl. **Zhang, K.** (2022), ApJ, 927, 229.
40. Calahan, J., Bergin, E., **Zhang, K.**, et al. (2021), ApJ, 908, 8.
39. Anderson, D., Blake, G., et al., incl. **Zhang, K.** (2021), ApJ, 909, 55.
38. Bosman, A., Alarcón, F., **Zhang, K.**, et al. (2021), ApJ, 910, 3.
37. Öberg, K., Guzmán, V., et al., incl. **Zhang, K.** (2021), ApJS, 257, 1.
36. Czekala, I., Loomis, R., et al., incl. **Zhang, K.** (2021), ApJS, 257, 2.
35. Law, C., Loomis, R., et al., incl. **Zhang, K.** (2021), ApJS, 257, 3.
34. Law, C., Teague, R., et al., incl. **Zhang, K.** (2021), ApJS, 257, 4.
33. Guzmán, V., Bergner, J., et al., incl. **Zhang, K.** (2021), ApJS, 257, 6.
32. Bosman, A., Alarcón, F., et al., incl. **Zhang, K.** (2021), ApJS, 257, 7.
31. Alarcón, F., Bosman, A., et al., incl. **Zhang, K.** (2021), ApJS, 257, 8.
30. Ilee, J., Walsh, C., et al., incl. **Zhang, K.** (2021), ApJS, 257, 9.
29. Cataldi, G., Yamato, Y., et al., incl. **Zhang, K.** (2021), ApJS, 257, 10.
28. Le Gal, R., Öberg, K., et al., incl. **Zhang, K.** (2021), ApJS, 257, 12.
27. Aikawa, Y., Cataldi, G., et al., incl. **Zhang, K.** (2021), ApJS, 257, 13.
26. Sierra, A., Pérez, L., **Zhang, K.**, et al. (2021), ApJS, 257, 14.
25. Bosman, A., Bergin, E., et al., incl. **Zhang, K.** (2021), ApJS, 257, 15.
24. Booth, A., Tabone, B., et al., incl. **Zhang, K.** (2021), ApJS, 257, 16.

23. Calahan, J., Bergin, E., **Zhang, K.**, et al. (2021), ApJS, 257, 17.
22. Teague, R., Bae, J., et al., incl. **Zhang, K.** (2021), ApJS, 257, 18.
21. Huang, J., Bergin, E., et al., incl. **Zhang, K.** (2021), ApJS, 257, 19.
20. Schwarz, K., Calahan, J., **Zhang, K.**, et al. (2021), ApJS, 257, 20.
19. Krijt, S., Bosman, A., **Zhang, K.**, et al. (2020), ApJ, 899, 134.
18. Alarcón, F., Teague, R., **Zhang, K.**, et al. (2020), ApJ, 905, 68.
17. Salyk, C., Lacy, J., et al., incl. **Zhang, K.** (2019), ApJ, 874, 24.
16. Schwarz, K., Bergin, E., et al., incl. **Zhang, K.** (2019), ApJ, 877, 131.
15. Anderson, D., Blake, G., et al., incl. **Zhang, K.** (2019), ApJ, 881, 127.
14. Schwarz, K., Bergin, E., et al., incl. **Zhang, K.** (2018), ApJ, 856, 85.
13. Rubinstein, A., Macías, E., et al., incl. **Zhang, K.** (2018), ApJ, 860, 7.
12. Gerdes, D., Sako, M., et al., incl. **Zhang, K.** (2017), ApJL, 839, L15.
11. Chen, X., Li, A., & **Zhang, K.** (2017), ApJ, 850, 104.
10. Blevins, S., Pontoppidan, K., et al., incl. **Zhang, K.** (2016), ApJ, 818, 22.
9. Schwarz, K., Bergin, E., et al., incl. **Zhang, K.** (2016), ApJ, 823, 91.
8. Bergin, E., Du, F., et al., incl. **Zhang, K.** (2016), ApJ, 831, 101.
7. Salyk, C., Lacy, J., et al., incl. **Zhang, K.** (2015), ApJL, 810, L24.
6. Salyk, C., Pontoppidan, K., et al., incl. **Zhang, K.** (2014), ApJ, 792, 68.
5. Jiang, B., Li, A., **Zhang, K.**, et al. (2014), IAU Symp., 297, 219.
4. Jiang, B., **Zhang, K.**, Li, A., et al. (2013), ApJ, 765, 72.
3. Bergin, E., Cleeves, L., et al., incl. **Zhang, K.** (2013), Nature, 493, 644.
2. Jiang, B., **Zhang, K.**, & Li, A. (2010), Earth Planets Space, 62, 105.
1. Jiang, B., **Zhang, K.**, & Li, A. (2005), ApJL, 630, L77.

Talks

Invited Talks at Conferences and Workshops

- Invited talk, Discs on the Exe conference, Exeter, UK 07/2026
  Protoplanetary structures and evolution
- Invited talk, 245th AAS meeting, Special Session: New perspectives on protoplanetary disks in the era of JWST and the ALMA Wideband Sensitivity Upgrade, Maryland, USA 01/2025
- Invited talk, Goldschmidt workshop "Exoplanets: Compositions, Mineralogy, Evolution", Chicago, USA 08/2024
  Chemistry of protoplanetary disks
- Invited talk, Origins of Solar Systems Gordon Research Conference, MA, USA 06/2023
  Gas Evolution of Protoplanetary disks
- Invited talk, "The Volatile Content of Planets that Form Early", Netherlands 11/2021

- Invited talk, "Planet-forming Disks: From Surveys to Answers", Netherlands 09/2021
- Invited talk, Community Science workshop of Origins Space Telescope 06/2019
- Invited talk, "Protoplanetary disk formation and evolution", Netherlands 07/2017
- Invited talk, "Comets: A new vision after Rosetta/Philae", France 11/2016

Colloquia and Seminars

- Colloquium, Department of Physics, University of Wisconsin-Madison, Wisconsin 10/2026
  Mapping the Evolution of Planet-Forming Disks: From Global Gas Reservoirs to Inner Disk Chemistry
- Seminar, Astrophysics group, Los Alamos National Laboratory, New Mexico 09/2026
  Mapping the Evolution of Planet-Forming Disks: From Global Gas Reservoirs to Inner Disk Chemistry
- Seminar, University of Michigan, Michigan 08/2026
  Mapping the Evolution of Planet-Forming Disks: From Global Gas Reservoirs to Inner Disk Chemistry
- Colloquium, University of Illinois Urbana-Champaign, Illinois 04/2025
  Gas evolution in protoplanetary disks
- Colloquium, University of Virginia, Virginia 03/2025
  Gas evolution in protoplanetary disks
- Colloquium, Center for Computational Astrophysics, New York 02/2025
  Gas evolution in protoplanetary disks
- Planet formation group seminar, Center for Computational Astrophysics, New York 02/2024
  Gas evolution in protoplanetary disks
- Seminar, AstroCheminar, online 01/2024
  Gas evolution in protoplanetary disks
- Colloquium, Iowa State University 10/2023
  Gas evolution in protoplanetary disks
- Seminar, JPL 09/2023
  Water in protoplanetary disks
- Seminar, UW-Madison Astronomy Monday Seminar 05/2022
- Colloquium, University of Chicago 04/2022
- Seminar, Origins of Life, Artificial Life & Astrobiology Seminar Series, UW-Madison 03/2022
- Seminar, University of Iowa, Astronomy/Space Physics Seminar 03/2021
- Seminar, MPIA planet formation group 01/2021
- Seminar, University of Arizona 11/2019
- Colloquium, Ohio State University 10/2019
- Seminar, Canadian Institute for Theoretical Astrophysics, Toronto, Canada 08/2019
- Colloquium, University of Florida 03/2019
- Colloquium, University of Wisconsin-Madison 02/2019
- Colloquium, University of Rochester 02/2019

- Colloquium, University of Wisconsin-Madison 03/2018
- Colloquium, University of Illinois at Urbana-Champaign 01/2018
- Colloquium, University of Michigan 09/2015

Contributed Talks

- AAS winter meeting, online 01/2021
- Astrochemical Frontiers, IAU online conference 07/2020
- Hubble Symposium 03/2019
- Exoplanet research conference in the Great Lakes area 12/2018
- "Astrochemistry: Past, Present, and Future", Pasadena, CA 07/2018
- Hubble Symposium 03/2018
- IAU Astrochemistry VII, Chile 03/2017
- NRAO TUNA lunch talk 02/2017
- "Young Solar Systems", Barcelona, Spain 04/2016
- CfA radio and geoastronomy division lunch talk, Chile 10/2014
- AAS winter meeting, Washington, D.C. 01/2014
- Yuk lunch seminar talk, CA 12/2013
- NRAO TUNA lunch talk, Charlottesville, VA 11/2012

Public Talks

- Science talk at the Space Place of the UW-Madison, Department of Astronomy 07/2022
- NASA Webb Space Telescope Community Event at Frankfort Public library district (virtual) 10/2021
- Earth 2.0, UW-Madison Astronomy Public talk series (virtual) 05/2021

Teaching and Mentoring

Teaching

- ASTRON 140, The Exoplanet Revolution 01/2026
- ASTRON 320, Physics of the Interstellar Medium, 09/2020, 09/2022, 09/2024
- ASTRON 910, Seminar in Astrophysics 09/2023
- ASTRON 103, The Evolving Universe: Stars, Galaxies, and Cosmology 01/2023
- ASTRON 340, Solar System Physics 01/2022, 01/2024, 01/2025
- ASTRON 700, Basic Astrophysics I - Radiation 09/2021
- AST 401 Exoplanets, University of Michigan, Substitute Teacher 02/2019
- AST 115 Introductory Astrobiology, University of Michigan, Substitute Teacher 09/2017–11/2017

- AST 115 New Discoveries in Astronomy, UMich, Guest Lecturer 09/2015, 02/2016
- AY 117 Statistics and Data Analysis, Caltech, Graduate Teaching Assistant 01/2012
- AY 125 High-Energy Astrophysics, Caltech, Graduate Teaching Assistant 01/2011
- AY 102 Physics of the Interstellar Medium, Caltech, Graduate Teaching Assistant 01/2010
- AY 123 Structure and Evolution of Stars, Caltech, Graduate Teaching Assistant 09/2010
- Introduction to FORTRAN, Beijing Normal University, Graduate Teaching Assistant 01/2006

Mentoring

Postdocs advised

- Eric Van Clepper, UW-Madison 09/2026–Present
  WiCOR Postdoctoral Fellowship
- Abygail Waggoner, UW-Madison 09/2024–Present
  NSF Astronomy and Astrophysics Postdoctoral Fellowship (2026–Present)
- Leon Trapman, UW-Madison 11/2020–10/2024

Graduate students advised

- Eshan Raul, UW-Madison 09/2024–09/2026
- Tayt Armitage, UW-Madison 09/2023–Present
  NSF Graduate Research Fellowship
- Estephani TorresVillanueva, UW-Madison 08/2021–Present
  NSF Graduate Research Fellowship
- Jenny Calahan, UMich, co-advised with Edwin Bergin 09/2018–07/2020
  NSF Graduate Research Fellowship; NASA FINESST Fellowship
- Felipe Alarcon Pena, UMich, co-advised with Edwin Bergin 09/2018–07/2020
- Christopher Merchantz, UMich, co-advised with Edwin Bergin 09/2017–06/2018

Undergraduate students advised

- Alonzo Ma, UW-Madison 01/2026–Present
- Nick Tallon, UW-Madison, co-advising with Abygail Waggoner 09/2024–Present
  2025 UW-Madison sophomore fellowship
  2025 Wisconsin Space Consortium undergraduate research award
- Carl Yao, UW-Madison 06/2024–Present
  2025 UW-Madison sophomore fellowship
- Nina Weichmann, UW-Madison 02/2023–05/2025
  2023 Wisconsin Space Consortium undergraduate research award
- Ella Chevalier, UW-Madison 09/2022–05/2025
  2023 UW-Madison sophomore fellowship
- Leopold Hutnik, UW-Madison, co-advised with Leon Trapman 10/2022–05/2024
- Vicki Braianova, UW-Madison 09/2021–12/2022
  2022 UW-Madison sophomore fellowship

- Anna Conly, UW-Madison REU student 06/2021–08/2021
- Olive Zhang, UW-Madison, co-advised with Leon Trapman 03/2021–04/2022
- Ben Capistrant, UW-Madison 09/2020–11/2021
- Matt Kalscheur, UW-Madison post-bachelor student 09/2020–08/2021
- Deryl Long, UMich, co-advised with Edwin Bergin 06/2019–08/2020
- Yuan Chen, UMich visiting student, co-advised with Edwin Bergin 09/2018–01/2019
- Maria Laura Ribeiro, Post bachelor student, co-advised with Edwin Bergin 09/2018–03/2019
- Tia Jin, UMich, co-advised with Edwin Bergin 09/2017–12/2017
- Jingyuan Li, Caltech, co-advised with Geoffrey Blake 06/2011–08/2011
- Stacy King, Caltech, co-advised with Geoffrey Blake 06/2010–08/2010

Service

Conference and Workshop Organization

- SOC member, *Missing Links in Planet Formation*, Taipei, Taiwan 11/2026
- SOC member, Great Lakes Exoplanet Area Meeting (GLEAM) 11/2025
- Co-organizer, JWST proposal workshop, University of Michigan 03/2020
- Organizer, UM Astronomy department colloquium series 06/2017–04/2018
- Organizer and lecturer, NRAO/ALMA proposal workshop, University of Michigan 03/2017

Proposal Review and Time Allocation

- JWST GO5 Review Panelist 02/2026
- JWST GO4 Review Panelist 02/2025
- NASA Decadal preparation science Review Panelist 07/2024
- NASA Exoplanet Research Program Review Panelist 07/2024
- JWST GO2 External Reviewer 03/2023
- ALMA Proposal Review Panelist 06/2021
- JWST GO1 Proposal Review Panelist 02/2021
- NASA proposal review Panelist 08/2019
- ALMA Proposal Review Panelist 06/2019
- External Reviewer for NASA FINESST program 03/2019
- Hubble Space Telescope Review Panelist 05/2018
- External Reviewer for NASA emerging world program 01/2017
- External Reviewer for NASA Astrobiology Institute 01/2017
- IRAM NOEMA/30m Time Allocation Committee, University of Michigan 09/2015–07/2020

Refereeing and External Thesis Committees

- Referee: Nature, Nature Astronomy, ApJ, ApJL, A&A, MNRAS Letters, PSS 01/2014–Present
- PhD thesis committee member for Jenny Calahan, University of Michigan 11/2020–2022

Service at UW-Madison

- Undergraduate faculty advisor, Astronomy Department 09/2022–Present
- Curriculum Committee member of Astronomy Department 09/2024–Present
- Colloquium organizer of Astronomy Department 09/2022–05/2024
- Committee of three for graduate students in Astronomy Department 09/2020–Present
- PhD thesis committee member for Dan Rybarczyk, Astronomy Department 05/2023
- Chair of Graduate Admission Committee, Astronomy Department 09/2021–04/2022
- Member of Prelim Committee, Astronomy Department 09/2021–04/2022
- PhD thesis committee member for William Thompson, Chemistry Department 10/2021–05/2024
- PhD thesis committee member for Logan Jones, Astronomy Department 07/2021
- Member of Graduate Admission Committee, Astronomy Department 09/2020–04/2021
- Member of Strategic plan steering Committee, Astronomy Department 09/2020–08/2021
- Telescope Time Allocation Committee, Astronomy Department 09/2020–Present

Technical Expertise

- **(Sub)mm interferometry:** ALMA, NOEMA, and VLA observation design and data reduction, continuum and spectral line
- **JWST mid-infrared spectroscopy:** MIRI/MRS observation design, data reduction, and molecular line analysis
- **Ground-based infrared spectroscopy:** Keck/NIRSPEC (20 nights) and Gemini/Michelle; spectro-astrometry and high dynamic range reduction
- **Disk modeling:** dust and line radiative transfer (rotational and rovibrational tracers); kinetic chemical evolution of disks